United International University

Department of Computer Science and Engineering

LAB REPORT

***Course Code*:** CSE 4326

***Course Title*:** Sessional based on ETE 4325

***Experiment No*:** 03

***Experiment Name*:** Familiarization with ESP32-based IoT systems

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Objective

1. To measure temperature and humidity using ESP32 and DHT11 sensor.
2. To display sensor data on Serial Monitor.
3. To send sensor data to a local XAMPP server.
4. To store the data in a MySQL database.
5. To develop a web interface for data visualization.

Theory:

Temperature and humidity monitoring is an essential part of environmental sensing systems used in applications such as weather stations, smart homes, agriculture, and industrial automation. In this task, an ESP32 microcontroller is used to interface with a DHT11 sensor to measure and display environmental conditions.

In this task, the ESP32 microcontroller is used to measure temperature and humidity using the DHT11 sensor. The DHT11 senses environmental conditions and sends digital data to the ESP32 through a single data pin. The ESP32 processes this data and displays the temperature (°C) and humidity (%) on the Serial Monitor.

This system demonstrates basic sensor interfacing, real-time data acquisition, and monitoring, which are fundamental concepts in embedded systems and IoT applications.

Components Required

- ESP32 Development Board
- DHT11 Sensor
- Jumper Wires
- Breadboard
- USB Cable
- Computer with Arduino IDE and XAMPP

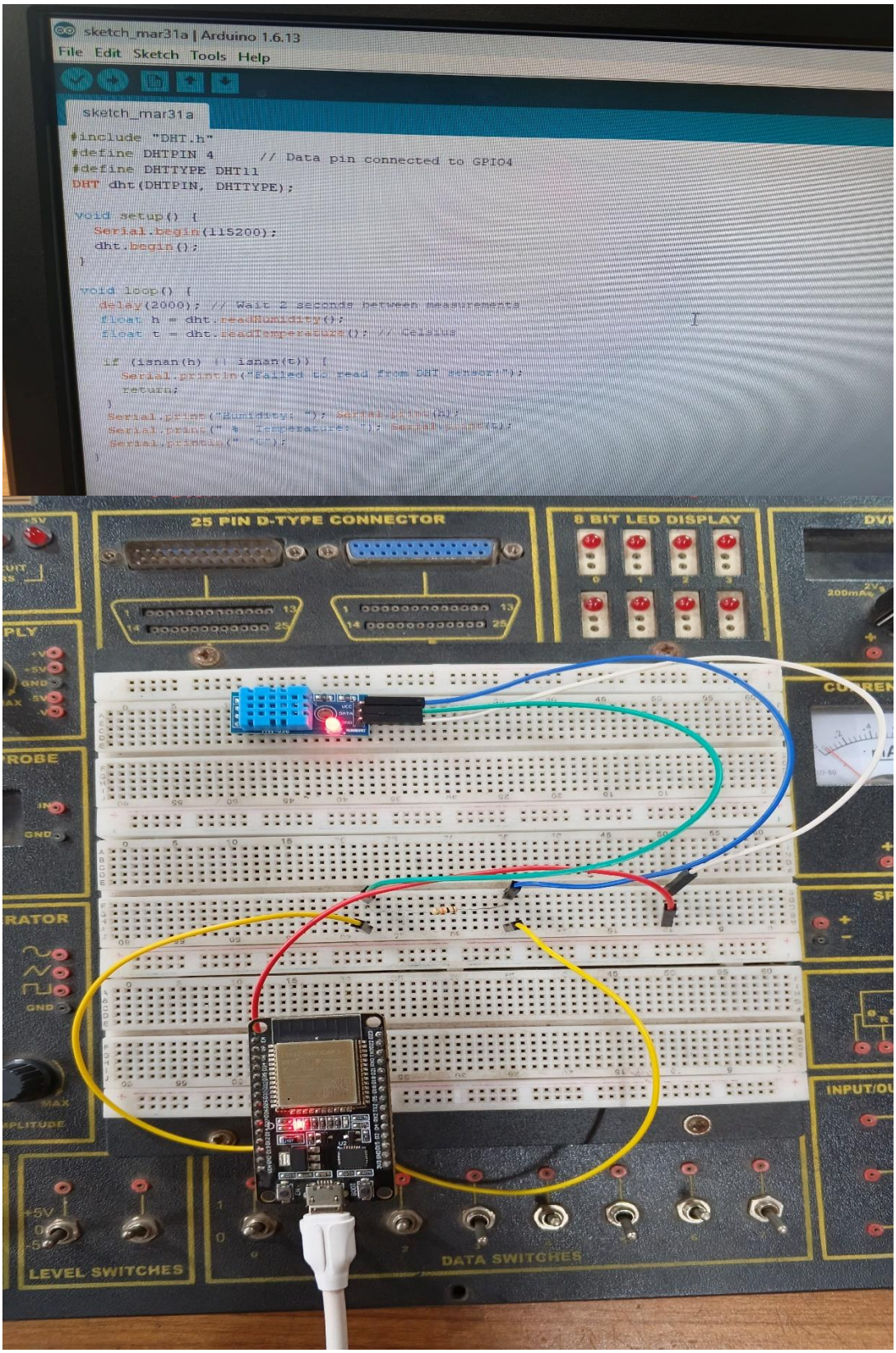
Task 1: Temperature and Humidity Monitoring

Circuit Diagram (Description)

- VCC of DHT11 → 3.3V (ESP32)
- GND of DHT11 → GND (ESP32)
- DATA pin → GPIO 4 (ESP32)
- Use a 10kΩ pull-up resistor between VCC and DATA (optional)

Program Code (ESP32)

```
#include "DHT.h"
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup() {
Serial.begin(115200);
dht.begin();
}
void loop() {
float humidity = dht.readHumidity();
float temperature = dht.readTemperature();
if (isnan(humidity) || isnan(temperature)) {
Serial.println("Failed to read from DHT sensor!");
return;
}
Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" °C");
Serial.print("Humidity: ");
Serial.print(humidity);
Serial.println(" %");
delay(2000);
}
```


sketch_mar31a | Arduino 1.6.13
File Edit Sketch Tools Help
sketch_mar31a
#include "DHT.h"
#define DHTPIN 4 // Data pin connected to GPIO4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup() {
Serial.begin(115200);
dht.begin();
}
void loop() {
delay(2000); // Wait 2 seconds between measurements
float h = dht.readHumidity();
float t = dht.readTemperature(); // Celsius
if (isnan(h) || isnan(t)) {
Serial.println("Failed to read from DHT sensor!");
return;
}
Serial.print("Humidity: "); Serial.print(h);
Serial.print(" % Temperature: "); Serial.print(t);
Serial.println(" °C");
}
25 PIN D-TYPE CONNECTOR
8 BIT LED DISPLAY
LEVEL SWITCHES
DATA SWITCHES

Demonstration Results

Example Serial Monitor Output:

Temperature: 29.00 °C
Humidity: 65.00 %

Task 2: IoT Data Logging using XAMPP Server

Step 1: Configure XAMPP

- Start Apache Server
- Start MySQL Server
- Open phpMyAdmin

Step 2: Create Database and Table

```
CREATE DATABASE sensor_db;

USE sensor_db;

CREATE TABLE data (
id INT AUTO_INCREMENT PRIMARY KEY,
temperature FLOAT,
humidity FLOAT,
timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Step 3: PHP Script (save_data.php)

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "sensor_db";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
```

```
die("Connection failed: " . $conn->connect_error);
}

$temperature = $_GET['temperature'];
$humidity = $_GET['humidity'];

$sql = "INSERT INTO data (temperature, humidity) VALUES ('$temperature',
'$humidity')";

if ($conn->query($sql) === TRUE) {
echo "Data inserted successfully";
} else {
echo "Error: " . $sql . "\n" . $conn->error;
}

$conn->close();?>
```

ESP32 Code to Send Data

```
#include <WiFi.h>
#include <HTTPClient.h>
#include "DHT.h"

const char* ssid = "DARKNESS";
const char* password = "tasin111";
const char* serverName = "http://localhost-3000/save_data.php";

#define DHTPIN 4
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {
Serial.begin(115200);
WiFi.begin(ssid, password);
dht.begin();

while (WiFi.status() != WL_CONNECTED) {
delay(1000);
Serial.println("Connecting...");
```

```
}
}

void loop() {
if (WiFi.status() == WL_CONNECTED) {
HTTPClient http;

float temperature = dht.readTemperature();
float humidity = dht.readHumidity();

String url = serverName;
url += "?temperature=" + String(temperature);
url += "&humidity=" + String(humidity);

http.begin(url);
int httpResponseCode = http.GET();

Serial.println(httpResponseCode);
http.end();
}

delay(5000);
}
```

Database Output (Example)

ID	Temperature	Humidity	Timestamp
1	29.0	65.0	2026-04-01 10:20:00

Task 3: Web-Based Sensor Data Visualization

PHP Code (display.php)

```
<?php
$conn = new mysqli("localhost", "root", "", "sensor_db");

$result = $conn->query("SELECT * FROM data ORDER BY id DESC");

while($row = $result->fetch_assoc()) {
```

```
echo "Temperature: " . $row['temperature'] . " °C - Humidity: " . $row['humidity'] .
" %<br>";
}
?>
```

HTML Webpage

```
<!DOCTYPE html>
<html>
<head>
<title>Sensor Data</title>
<meta http-equiv="refresh" content="5">

<style>
body{
   font-family: Arial;
   background:#f4f4f4;
   text-align:center;
   padding:40px;
}

h2{
   color:#333;
}

.data{
   background:#fff;
   padding:20px;
   border-radius:10px;
   display:inline-block;
   box-shadow:0 0 10px rgba(0,0,0,0.1);
}
</style>

</head>
<body>

<h2>Temperature and Humidity Data</h2>

<div class="data">
   <?php include 'display.php'; ?>
```

```
</div>

</body>
</html>
```

Conclusion:

The project successfully demonstrates real-time temperature and humidity monitoring using ESP32 and DHT11. Data is transmitted to a local server, stored in a database, and visualized through a web interface. This system can be extended for smart home and industrial monitoring applications.